

Affective Tagging Theory (ATT): A Mathematical Architecture Unifying Informational Topology and Global Workspace Dynamics

Élisée Kadima Nsabua

Independent Researcher

2026

Abstract

This manuscript introduces the formal mathematical architecture of Affective Tagging Theory (ATT), a multi-scale theoretical framework designed to bridge the gap between biological neural networks and subjective experience (qualia). By uniting principles from Integrated Information Theory (IIT), Global Workspace Theory (GWT), and predictive coding, ATT formalizes how the brain utilizes dualistic chemical and synaptic valence tags ($\tau^\pm$) to track behavioral deviations from dynamic reference manifolds. We demonstrate that consciousness is a time-bound, cumulative volume (C) driven by the multiplicative interaction between a high-contrast workspace gating mechanism and an irreducible valential-informational core. This work provides a rigorous calculus for computational psychiatry, offering precise mathematical resolutions to classic phenomenological paradoxes such as constant sensory suffering, localized emotional valence propagation, and macro-scale emotional emergence.

1 Introduction & Theoretical Framework

The physical mechanisms underlying the generation of subjective, qualitative experience—what philosophy terms the “Hard Problem” of consciousness—remain the central battleground of modern cognitive neuroscience. While contemporary neuroimaging can map neural correlates of perception with high spatial fidelity, classical neuroscience lacks a unified mathematical language capable of translating raw metabolic activity and cellular firing rates into the dense, unified, and emotionally charged geometry of human awareness.

Affective Tagging Theory (ATT) addresses this explanatory gap by proposing that consciousness does not arise from isolated, static neural nodes. Instead, subjective experience is generated through a continuous process of contrastive affective tagging. The brain acts as a dynamic differential engine, constantly comparing real-time neural state vectors (x) against historical, predictive baselines (B_{hist}). The resulting architectural deviations are instantly marked by an irreducible paired set of positive and negative valential markers ($\tau^{\pm}$).

Crucially, ATT argues that human emotion is not a vague, unquantifiable epiphenomenon, but rather the very metric the system uses to calculate metabolic strain and drive topological network reconfiguration. By formalizing these affective markers through vector calculus, information theory, and statistical mechanics, this manuscript establishes a rigorous, evaluable framework where qualia are treated not as abstract philosophical mysteries, but as predictable, geometric consequences of an integrated informational system. The following sections outline the twenty-seven foundational equations that govern this subjective space.

Phase 1: Micro-Foundations (The Building Blocks)

Equation 1: The Mechanistic Foundation (Total Consciousness)

Modern neuroscience has moved beyond philosophical debate to focus on the mechanistic foundations of human consciousness. The Affective Tagging Theory (ATT) proposes that conscious experience (qualia) does not arise from isolated neural activity. Instead, it emerges through contrastive affective tagging—a process where the brain marks neural patterns based on their emotional significance relative to dynamic baselines.

This foundational mechanism can be translated into a rigorous calculus tracking the temporal evolution of conscious volume, which maps directly to the formal relation:

$$C = \int_{t_0}^{t_f} G(\nabla A) \cdot f(\tau^\pm, \Phi) dt \quad (1)$$

Where the active variables and parameters are defined verbatim from the master architectural framework:

- C : Total Consciousness — Cumulative conscious volume over a temporal window $[t_0, t_f]$.
- G : Global Affective Workspace — The “arena” where high- Φ configurations outcompete noise for conscious broadcast.
- ∇A : Unified Affective Delta — Denotes the total directional gradient or movement away from the baseline reference manifold B_{hist} .
- f : The irreducible valential-informational core function mapping the structural architecture of the affective system.
- $\tau^\pm$: Dualistic Logic (Affective Tags) — Pairs positive and negative tags as a single irreducible unit. It defines a non-decomposable contrastive pair where the tag and its deviator are mutually dependent for valential definition.

- Φ : Integrated Information — Represents the degree of irreducible interdependence between the systemic positive and negative affective tag configurations.

Mathematically, this cumulative conscious volume C is formalized as a definite integral over a temporal window $[t_0, t_f]$, demonstrating that awareness is an inherently dynamic, time-bound accumulation rather than a collection of frozen, static snapshots. Within this calculus, the total magnitude of experience is determined by the multiplicative interaction between a global workspace gating function $G(\nabla A)$ —which measures how intensely a trajectory-shifting contrastive delta (∇A) recruits the global workspace—and an irreducible valential-informational core $f(\tau^\pm, \Phi)$, which synthesizes the integrated interdependence (Φ) of coupled positive and negative affective tags ($\tau^\pm$). Because these two components are bound multiplicatively, total consciousness collapses to zero if either side of the equation fails; a deeply integrated valential signature (f) that lacks the high-contrast velocity required to ignite the workspace ($G \rightarrow 0$) remains trapped as an ambient, subconscious mood, whereas a massive global workspace broadcast that lacks structured, integrated emotional meaning ($f \rightarrow 0$) merely dissipates as uncoordinated metabolic noise.

Phenomenological Anchor: Consciousness sparks the moment a change disrupts what is normal (B_0 or B_{hist}) into something same but different. This creates an immediate contrast that is globally broadcasted (G) through the mind. This broadcast is not empty data; it is the instant, irreducible (Φ) knowing of what to do versus what not to do, where to go versus where not to go, and what is now versus what should be. Each distinct contrast forms an unbreakable, foundational piece of raw awareness.

Equation 2: Definition of Subjective Genesis Space

The Mechanistic Foundation In this model, neuromodulators drive synaptic plasticity to encode the importance of events directly into the brain’s physical structure. These “affective tags” (τ^+ , τ^-) bias neural activity toward either

positive or negative states. Subjective feeling is the result of the brain’s dynamic constant evaluation of the velocity of neural shifts and the divergence between current states and historical manifolds; it is the moment-to-moment dynamic “readout” of how neural information interacts within an irreducible configuration.

This multi-dimensional computational pressure and its intersection with structural memory map directly to the formal relation:

$$S = G \cdot \sum_{i=1}^n \sigma_i \left[\left(\frac{dx_i}{dt} \right) + (x_i - B_{\text{hist},i}) \right] \quad (2)$$

Where the instantaneous state evaluation within the pre-conscious architecture is defined by the following local parameters:

- S : Subjective Genesis Space — representing the instantaneous phase space where physical neural velocity and cognitive deviations intersect.
- G : Global Affective Workspace — the competitive arena where high-urgency configurations outcompete noise for conscious broadcast.
- σ_i : Synaptic Sensitivity (Gain) — acting as the directional scaling coefficient or functional volume weight of the i -th neural population.
- x : Neural Configuration (State) — Represents the current vector position of the active neural assembly in state-space.
- $B_{\text{hist},i}$: Historical Manifold of the i -th population, the moving reference baseline tracking long-term neural state expectations.
- n : The complete operational boundary integer of active feature nodes inside the local network.

The apparent “dimensional leak” in Equation 2—adding velocity ($\frac{dx}{dt}$) to displacement ($x - B_{\text{hist}}$)—is not a glitch, but rather a formalization of the **Law of Unitary Urgency**. In the pre-conscious staging ground of the brain, the system does not distinguish between “how fast a signal is changing” and “how far it has strayed from the baseline”; instead, it flattens these variables

into a single, unitless metric of **Computational Pressure**. By utilizing σ (Synaptic Gain) as a filter, the equation mathematically simulates how the brain de-prioritizes noise. The G (Global Workspace) factor then acts as the “Selection Operator,” determining which of these competing parallel streams possess enough total urgency to be broadcasted to the wider network. Without this “flattening,” the theory would fail to explain how a slow, chronic cognitive divergence (like grief) can compete for attention against a sharp, instantaneous sensory flux (like a pinprick).

The link between Equation 2 and Equation 3 represents the transition from **Parallel Competition** to **Singular Realization**. While Equation 2 (S) describes the “Genesis Space”—a messy, high-entropy arena where thousands of potential deltas vie for dominance—Equation 3 (q_1) represents the “Collapse” of that space into the first polarized quale. Mathematically, Equation 3 acts as the limit-break where the winner of the competition in S crosses the threshold of the Global Workspace to become a felt experience. In this pipeline, S is the **substrate of possibility**, and q_1 is the **event of awareness**. By linking them this way, the theory elegantly explains why we aren’t overwhelmed by every minor neural fluctuation: unless the sum of urgency in Equation 2 is sufficient to break the “limit” in Equation 3, the information remains sub-threshold noise, never achieving the status of awareness.”

The Dual-Baseline Architecture and the Temporal Derivative

The genesis of conscious feeling is strictly dependent on a directional movement away from a sliding stasis. This is governed by two distinct reference points:

- **The Immediate Baseline (B_0):** A Sliding Physiological Anchor representing the sensory “zero-point” or assimilated stasis of the present moment.
- **The Historical Baseline (B_{hist}):** A Fixed Topological Ideal or preserved historical manifold of a prior equilibrium (e.g., the remembered

comfort of Earth’s 1x gravity).

Phenomenological Anchor: Subjectively, Equation 2 clarifies how an enduring emotional state like deep grief or existential dread can demand conscious attention just as intensely as a sudden physical crisis like a needle prick. Even when the physical sensory velocity ($\frac{dx_i}{dt}$) of a long-term state approaches zero, its persistent, structural divergence from our historical baseline expectations ($x_i - B_{\text{hist},i}$), multiplied by high localized synaptic gain (σ_i), builds a tremendous aggregate computational pressure. This mathematical flattening explains why historical psychological pain can successfully outcompete background neural noise and occupy the central focus of the global conscious workspace (G) alongside acute incoming sensory shocks.

Equation 3: The Law of Affective Contrast (LAC)

Phenomenal valence is a directional vector of deviation gated by temporal velocity and predictive topology; the first quale emerges from the first movement away from a baseline equilibrium.

This thresholded emergence can be explicitly formalized by tracking the instantaneous geometric deviation against the activation gate of the affective workspace:

$$q_1 = \Delta|x - B_0| \cdot [|\nabla A| > 0] \quad (3)$$

Where the operational behavior of this comparative gateway is governed by the following active parameters:

- q_1 : The primary quale — representing the discrete unit of subjective experience generated by a non-zero affective delta.
- x : The neural configuration state — marking the current vector position of the neural assembly in state-space.
- B_0 : The static baseline — defining the reference manifold or “zero-point” from which all metric deviations are measured.

- ∇A : Unified Affective Delta — Denotes the total directional gradient or movement away from the baseline reference manifold B_{hist} .

Equation 3 serves as the mathematical gatekeeper for the Law of Affective Contrast (LAC), asserting that conscious experience—specifically the “first quale” (q_1)—is not a product of raw intensity, but of directional change. Under the Static Pain Paradox, a constant physiological burden, such as the 7x gravity mentioned in the thought experiment, is effectively absorbed into the baseline (B_0), remaining phenomenally silent. It is only when a Directional Delta (∇A) occurs that the system breaks its affective neutrality. This movement creates a valential bridge, retroactively polarizing the previous stasis as either “bad” or “good” based on whether the shift represents metabolic relief or further homeostatic strain. Technically, the equation achieves this through the use of the Iverson bracket $[|\nabla A| > 0]$, which acts as a binary switch for consciousness. If there is no rate of change in the neural configuration relative to the baseline, the bracket evaluates to zero, mathematically silencing the magnitude of the stimulus regardless of its objective physical scale. However, the moment a non-zero “velocity” is detected, the bracket evaluates to one, allowing the magnitude of the shift ($\Delta|x - B_0|$) to be released into the Global Affective Workspace. This ensures that the theory remains parsimonious: it doesn’t require constant cognitive energy to “feel” a static state; instead, it optimizes the brain’s comparative engine to respond only to the deviations that mandate a survival-based behavioral adaptation.

Phenomenological Anchor: Without the Delta, the system enters a state of “Information Heat Death.” You could experience the most intense physical pressure in the universe, but if it is perfectly uniform and eternal, it becomes functionally equivalent to nothingness—the objective pressure is present, but you cannot assign a valence to it because it has become your normal state. The brain is not a meter; it is a comparator.

Equation 4a: Valential Coupling Information

Consciousness is a gradient of integration where meaning is an irreducible Φ -value generated by Contrastive Pairs ($\tau^\pm$); because the tag and its deviator are mutually dependent for valential definition, they form a single, non-decomposable unit of phenomenal tone.

This underlying non-decomposable contrastive unity maps directly to the formal relation:

$$I_\tau(\tau^+; \tau^-) = H(\tau^+) + H(\tau^-) - H(\tau^+, \tau^-) \quad (4a)$$

Where the informational behavior of the contrastive pair is defined by the following active variables and parameters:

- **I_τ** : Mathematically defines the interdependence of your τ^+ and τ^- tags to calculate the actual value of Φ .
- τ^+ : Positive Affective Tag — Represents the localized density of positive valential markers across the active network or individual node.
- τ^- : Negative Affective Tag — Represents the localized density of aversive or negative valential markers across the active network or individual node.
- **H** : Represents the complexity of a neural state; used to ensure Φ is non-decomposable (The Law of Irreducible Meaning).
- Ω_{st} : Stable Configuration — representing a global or localized state of positive valential dominance where (τ^+) tags outnumber or overpower (τ^-) tag that generates systemic geometric equilibrium manifesting phenomenally as the quale of joy, warmth, or attachment.
- Ω_{unst} : Unstable Configuration — representing a global or localized state of negative valential dominance where (τ^-) tags outnumber or overpower (τ^+) tag that generates systemic geometric tension manifesting phenomenally as the quale of pain, dislike, or hate

Equation 4a formalizes **Valential Coupling Information** (I_τ), which measures the shared informational variance or mutual information between the positive (τ^+) and negative (τ^-) valential tag distributions within the neural architecture. Instead of treating these tags as static, isolated markers, this equation utilizes marginal and joint entropies to capture the dynamic cross-talk and structural coupling between competing network states.

This shared information represents the relational “binding energy” of the system; it explains how an intrinsically neutral stimulus—such as the mental representation of an anticipated antidote—can structurally couple with an active, high-entropy pain network based entirely on its contextual relationship to the current state. **The antidote doesn’t have a fixed, hardcoded positive value on its own—its positive density profile is entirely generated by the context of the active pain network it interacts with.**

This informational coupling is the precise mechanism that drives the macroscopic phase transition from an unstable topological configuration to a stable one ($\Omega_{\text{unst}} \implies \Omega_{\text{st}}$). When the global workspace is trapped in a state of high aversion, the mutual information tracking channel (I_τ) allows the positive valential tags triggered by the antidote’s concept to precisely target and interlock with the active pain assembly. This sudden influx of positive tags mathematically reconfigures the valential summation of the network, shifting its geometric landscape to overpower the negative tag population. Equation 4a therefore provides the exact formal foundation for how cognitive context alters global network topology, collapsing a high-tension state of suffering into a stable, balanced equilibrium.

Equation 4b: Integrated Information (Φ) as Interdependence

The foundation of any conscious experience is Contrast. Feeling arises only when the brain detects a difference between a current neural state (x) and established baselines. This requires both positive and negative signals to work together as a unified whole, manifesting as in a high degree of Inte-

grated Information (Φ) where they become interdependent on one another for valential meaning.

This systemic requirement for co-dependent contrast can be formalized mathematically by subtracting the information of the partitioned parts from the integrated whole:

$$\Phi = I(\tau^+(x, B_{hist}); \tau^-(x, B_{hist})) - \sum_{i=1}^n I(\tau_i^+(x_i, B_{hist,i}); \tau_i^-(x_i, B_{hist,i})) \quad (4b)$$

Where the behavior of this informational field is governed by the interaction of the following local variables and parameters:

- Φ : Integrated Information — Represents the degree of irreducible interdependence between the systemic positive and negative affective configurations.
- I : Mutual Information — Measures the statistical dependence and informational cross-talk between opposing valential tags.
- τ^+ : Positive Affective Tag — Represents the localized density of positive valential markers across the active network or individual node.
- τ^- : Negative Affective Tag — Represents the localized density of aversive or negative valential markers across the active network or individual node.
- x : Neural Configuration (State) — Represents the current vector position of the active neural assembly in state-space.
- B_{hist} : Historical Manifold — The moving reference baseline tracking long-term neural state expectations.
- i : Node Index — Identifies individual localized channels or sub-networks within the broader partitioned topology.

- n : Node Boundary — An integer representing the total number of distinct parts or localized sub-networks analyzed across the system partition.

Equation 4b formalizes integrated information (Φ) as the mathematical proof of an irreducible, valential conscious unit. Crucially, the calculation shifts away from cold, neutral data vectors and applies itself directly to the contrastive affective tags (τ^+ and τ^-), which are explicitly parameterized as functions of the current state deviation (x) and the historical baseline manifold (B_{hist}). The first term, $I(\tau^+(x, B_{hist}); \tau^-(x, B_{hist}))$, quantifies the total mutual information—the systemic co-dependence—between the positive and negative poles generated by this interaction. By subtracting the second term, $\sum_{i=1}^n I(\tau_i^+(x_i, B_{hist,i}); \tau_i^-(x_i, B_{hist,i}))$, which calculates the sum of valential information if the network’s local nodes were completely partitioned, the equation isolates the precise amount of affective meaning lost upon reduction. If $\Phi > 0$, it proves that the system’s hedonic or aversive tone cannot be broken down into independent neural tracks, establishing consciousness as a unified gradient of integration.

Phenomenological Anchor: Phenomenologically, this mathematical architecture perfectly captures the mechanism of Long-Term potentiation (LTP) consolidation, where the historical baseline and the current deviator are welded into a single, non-decomposable unit of phenomenal tone. Within this framework, the state deviation x and the historical baseline B_{hist} do not possess inherent, independent emotional values; instead, they act as an inseparable contrastive pair that mutually polarize one another into τ^+ and τ^- in real-time. The baseline is only experienced as a state of relief or dread because it fires in direct opposition to the current deviation x , which in turn derives its specific aversive or positive intensity from its distance from the zero-point baseline. Because the positive tag and the negative tag are utterly dependent on this comparative relationship for their valential definition, they cannot be split into separate neural processing tracks without destroying the meaning of experience itself. They exist as an irreducible Φ -value, forcing the mathematics to explain itself as a single, unified flash of subjective meaning.

Phase 2: The Delta Architecture & The Static Pain Solution

Equation 5: The Zero-Delta Constraint (Genesis Paradox)

The Static Pain Paradox Resolution Consider a human born on a planet with gravity seven times heavier than Earth's. Their bodies would be in a constant state of immense physical strain, but because this state is the only reality they have ever known, their brains utilize the Comparative Engine of the Prefrontal Cortex (PFC) to attempt a comparative mapping of the current state. Initially, the 7x burden is an unpolarized baseline (B_0). Their brains would lack a lighter point of reference to tag the strain as “negative” or “positive.” This constant, non-changing physical burden would simply be their physiological and sensory baseline (B_0), resulting in a state of affective neutrality because there is no existing reference point to compare the current state against.

This state of absolute affective neutrality arising from static systemic strain maps directly to the formal baseline boundary relation:

$$\text{If } |\nabla A_s| + |\nabla A_c| = 0 \implies \Delta q = 0 \quad (5)$$

Where the physical boundary conditions are governed by the interaction of the following local operational variables:

- ∇A_s : Sensory Delta — Represents the instantaneous rate of change in physical neural input relative to the immediate baseline (B_0).
- ∇A_c : Cognitive Delta — Represents the spatial divergence between the current neural configuration state (x) and the long-term historical manifold (B_{hist}).
- Δq : Discrete Qualia Update — Represents the change, initialization, or localized shift in phenomenal experience over an evaluation step.

The Zero-Delta Constraint formalizes the system's structural boundary conditions, serving as an informational trapdoor where the total absence of affective velocity prevents a reset of the conscious baseline. By establishing that a complete lack of sensory and cognitive deltas ($|\nabla A_s| + |\nabla A_c| = 0$) forces the discrete qualia update to zero ($\Delta q = 0$), the framework introduces a fundamental structural asymmetry: while kinetic mental motion is strictly mandated to initially polarize and carve a new valential coordinate, zero ongoing motion is required to sustain a consolidated state. Once an intense state of qualia is brought online, a sudden collapse in trajectory leaves the system captured by its own historical inertia. The network is structurally blocked from resetting back to a true absolute zero baseline in a vacuum; it remains anchored to its existing valential coordinates until an explicit retroactive counter-force or subsequent delta arrives to displace it.

Phenomenological Anchor: If there is no gap between “What is” and “What could be,” there is no value.

- The 7x Paradox proves that “Badness” (τ^-) and “Goodness” (τ^+) are not properties of the universe; they are coordinates in a map of change.

Equation 6: Retroactive Polarization (6.5x Gravity Shift)

However, if gravity decreases to 6.5x, the brain identifies a reduction in homeostatic load and metabolic strain. This triggers a Retroactive Polarization: the current 6.5x state is assigned a positive τ^+ tag, while the previously neutral 7x state is functionally re-indexed as a negative τ^- reference point. This process creates the first Valential Bridge, where the 7x state only acquires its "badness" once the movement toward 6.5x provides the necessary contrast to reveal the metabolic cost of the original stasis.

Conversely, if gravity increases to 7.5x, the "velocity" of that change triggers the first negative tag (τ^-), retroactively defining the 7x state as a comparative τ^+ baseline. This transition allows the brain to finally categorize its surroundings. This dynamic re-indexing of relative affective states can be formally expressed as a piecewise vector mapping function based on differential

systemic strain, which maps directly to the formal relation:

$$\begin{bmatrix} \tau_{t_0}(x) \\ \tau_{t_1}(x) \end{bmatrix} = \begin{cases} \begin{bmatrix} \tau^+ \\ \tau^- \end{bmatrix}, & \text{if } M(x(t_0, B_{hist})) < M(x(t_1)) \\ \begin{bmatrix} \tau^- \\ \tau^+ \end{bmatrix}, & \text{if } M(x(t_0, B_{hist})) > M(x(t_1)) \end{cases} \quad (6)$$

Where the behavioral mechanics of this retroactive assignment are governed by the following local active variables and parameters:

- $\tau_{t_0}(x)$: Initial Valential State — The historic valential tag configuration evaluated at time t_0 .
- $\tau_{t_1}(x)$: Subsequent Valential State — The active valential tag configuration evaluated at time t_1 .
- τ^+ : Positive Affective Tag — Represents the density of positive affective markers across a network.
- τ^- : Negative Affective Tag — Represents the density of aversive markers across a network.
- $M(x)$: State Evaluation Metric (Strain / Cost Function) — A scalar metric mapping sensory intensity and metabolic strain for a given configuration state x .
- x : Neural Configuration (State) — Represents the current vector position of the neural assembly in state-space.
- B_{hist} : Historical Manifold — The moving reference baseline tracking long-term neural state expectations.

This equation structure formalizes valence not as an intrinsic, static property of an isolated neural state, but as a relational value dynamically assigned through prefrontal-mediated historical comparison. By framing the system as a joint piecewise vector function dependent on the strain metric $M(x)$, the math explicitly mandates that the affective tags at time t_0 and t_1 exist

in a state of reciprocal tension. The baseline state $\tau_{t_0}(x)$ cannot define its own valential polarity in a vacuum; instead, its tag is retroactively calculated only after a subsequent sensory or cognitive deviator arrives at t_1 , creating a formal mathematical dependency where the affective memory of the past is fundamentally color-coded by the trajectory of the present.

Phenomenological Anchor: Phenomenologically, this architecture perfectly mirrors the psychological realities of relative suffering and relief by prioritizing the direction of change over absolute intensity values. When the metabolic or sensory strain of the historical baseline $M(x(t_0, B_{hist}))$ is lower than that of the current deviator $M(x(t_1))$, the mathematical logic automatically crowns the past as a positive haven (τ^+), which simultaneously casts the current deviation as an aversive state (τ^-) of heightened distress. Conversely, if the deviator manages to drop below the baseline’s strain, the mechanism cleanly flips the polarity: the current state claims the positive affective tag (τ^+), retroactively recontextualizing the historical baseline as a negative state (τ^-) in hindsight. This elegant, symmetrical dependency guarantees that the network’s global landscape remains highly fluid, tracking the delta of homeostatic drift rather than getting trapped in static equilibrium.

Equation 7: The Reference Limit (Intensity vs. Delta)

The Static Pain Paradox: A perfectly steady pain signal would fail to produce a negative feeling. Because our aversive systems are tuned to detect deviations, a signal with zero variance provides no contrast for the brain to latch onto, no matter its intensity. This is because the brain would have no “low pain or high” (differential valential coordinates) “state” to compare the current state to. This thought experiment is critical because a perfectly static pain does not exist in the biological landscape. Even chronic pain is dynamic—it fluctuates, throbs, or is intermittent. A tag is created only after a stimulus increase or decrease. This shift causes the brain to retroactively tag the current and now-absent stimulus as “negative” or “positive,” whether it is an instant tagging or late consolidation.

This fundamental constraint of static intensity can be expressed mathematically through the following limit function, which maps directly to the formal relation:

$$\lim_{M(x) \rightarrow \infty} q(M(x)) \Big|_{\nabla A=0} = 0 \quad (7)$$

Where the behavioral boundaries of the experience are defined by the following active parameters and variables:

- $M(x)$: State Evaluation Metric (Strain / Cost Function) — A scalar metric mapping sensory intensity and metabolic strain for a given configuration state x .
- q : Affective Quale — Represents the discrete unit of phenomenal experience or the resulting strength of the qualitative readout.
- ∇A : Unified Affective Delta — Denotes the total directional gradient or movement away from the baseline reference manifold B_{hist} .

The Genesis Blindness Limit establishes the absolute requirement of an initial gradient for valential awakening, serving as the necessary counterpart to the Zero-Delta Constraint. This limit dictates that within an unprimed or native system, the state evaluation metric ($M(x)$)—no matter how macroscopically massive—fails to generate an affective quale ($q = 0$) if it is introduced as a perfectly static baseline devoid of initial deviation ($\nabla A = 0$). Phenomenologically, this accounts for the “Hell from Birth” condition: an organism born into a continuous state of extreme physiological or metabolic strain possesses no lighter coordinate to frame its reality as “suffering,” nor a heavier one to frame it as “relief.” The intense physical burden remains a completely unpolarized, affectively neutral baseline up to the point of biological destruction or cellular collapse. Consequently, while Equation 7 establishes a stasis boundary law—proving that a system devoid of kinetic velocity remains locked within its existing valential coordinates rather than resetting to an absolute zero baseline—Equation 8 demonstrates that raw intensity alone cannot ignite the initial valential coordinates of consciousness without a kinetic catalyst.

Phenomenologically, this architecture dictates that a sensory state completely stripped of dynamic variance becomes entirely imperceptible to our emotional architecture. This Reference Limit is fundamentally anesthetizing: it states that even if the state evaluation metric ($M(x)$) approaches infinity, if the unified affective delta (∇A) is zero, the resulting phenomenal quale (q) collapses to zero. This mathematical rule explains why chronic pain is so insidious—it throbs. If a pain signal lacked variance and the underlying neural assemblies fired at a perfectly constant, unvarying rate, the cognitive workspace would eventually treat the trauma exactly like the perpetual hum of a background fan, filtering it out of conscious awareness entirely. Ultimately, this highlights a profound systemic power dynamic: raw physical intensity alone does not dictate suffering; rather, the dynamic directional delta (∇A) holds absolute dominion over the generation of distress and relief.

Equation 8a: The Unified Affective Delta (∇A)

The Counterfactual Bridge ($\hat{\Delta}$)

While internal sensory shifts provide a biological delta (∇A_s), the Prefrontal Cortex provides a Cognitive Delta (∇A_c). (Based on the 7x gravity paradox) By observing another agent moving with ease, the PFC performs a comparative analysis, transforming a previously neutral baseline (B_0) into a polarized contrast configuration. Utilizing the B_{hist} manifold via the Counterfactual Bridge ($\hat{\Delta}$), the PFC maintains the divergence between the current state (x) and B_{hist} , explaining why an Earth-born human continues to suffer in 7x gravity; the memory of ease (B_{hist}) maintains the delta, ensuring the experience remains "visible" to consciousness as a motivator for change. This dual-source variance vector is mathematically unified by mapping real-time somatic shifts alongside amplified cognitive tracking, which maps directly to the formal relation:

$$\nabla A = \underbrace{\Delta x}_{\nabla A_s} + \underbrace{\sigma \odot (x - B_{hist})}_{\nabla A_c} \quad (8a)$$

where $\hat{\Delta} = x - B_{hist}$

Where the localized behavioral dynamics are defined by the interaction of the following operational variables and parameters:

- ∇A : Unified Affective Delta — Denotes the total directional gradient or movement away from the baseline reference manifold B_{hist} .
- Δx : State Displacement Vector — The real-time physical shift or velocity of the active neural population.
- ∇A_s : Sensory Delta — Represents the instantaneous rate of change in physical neural input relative to the immediate baseline (B_0).
- ∇A_c : Cognitive Delta — Represents the spatial divergence between the current neural configuration state (x) and the long-term historical manifold (B_{hist}).
- $\hat{\Delta}$: Counterfactual Bridge — Represents the simulated state-space divergence vector ($x - B_{hist}$) generated by tracking an expectation or memory baseline not physically present.
- σ : Synaptic Sensitivity (Gain) — Represents the weight or gain of a neural population; explains how repeated firing or Hebbian plasticity “overpowers” other signals regardless of raw tag count.
- x : Neural Configuration (State) — The current vector position of the active neural assembly in state-space.
- B_{hist} : Historical Manifold — The moving reference baseline tracking long-term neural state expectations.

The generation phase, formalized in Equation 8(a), captures the immediate structural collision of a neural event before any learning occurs. It synthesizes raw sensory velocity (∇A_s) with the cognitive delta (∇A_c), which measures how far the current neural state (x) has drifted from the historical manifold (B_{hist}) under the current element-wise multiplier of synaptic gain (σ). This combined vector represents the raw, real-time salience of the experience, providing the energetic momentum required to pierce through neural noise and capture focus within the Global Affective Workspace (G).

Phenomenological Anchor: Phenomenologically, this architecture explains why suffering can persist even in a completely static physical environment. While an individual standing perfectly still under an oppressive 7x gravity load lacks any real-time sensory displacement ($\nabla A_s = 0$), they do not become numb to their plight. Instead, the prefrontal cortex actively summons an alternate, counterfactual baseline of 1x gravity from earth stored deep within the historical manifold (B_{hist}). The ongoing mathematical divergence between their current structural reality (x) and this phantom memory of ease ($x - B_{hist}$), heavily scaled by active synaptic gain (σ), yields a persistent cognitive delta (∇A_c). This guarantees that the unified affective delta (∇A) remains wide open, generating a continuous stream of conscious distress fueled entirely by the contrast between what is and what was.

Equation 8b: The Chronological Integration Loop (Dual-Action System Update)

This downstream chronological loop and its dual-action system update can be formalized as, which maps directly to the formal relation:

$$\begin{cases} \Delta\Phi = \lambda \cdot I(\tau^+, \tau^-) \Big|_{\nabla A} \\ B_{hist}(t+1) = B_{hist}(t) + \lambda \cdot \nabla A_c \end{cases} \quad (8b)$$

Where the temporal updating mechanics are governed by the interaction of the following local active variables and parameters:

- $\Delta\Phi$: Integrated Information Variance — Represents the discrete change or accumulation of irreducible, interdependent valential information.
- λ : Teaching Signal (LTP/LTD) — Represents the update rule or learning rate triggered by the vividness and contrast of the quale.
- I : Mutual Information — Quantifies the statistical co-dependence between the opposing positive and negative poles.
- τ^+ : Positive Affective Tag — Represents the density of positive affective markers across the network.

- τ^- : Negative Affective Tag — Represents the density of aversive markers across the network.
- ∇A : Unified Affective Delta — Denotes the total directional gradient or movement away from the baseline reference manifold B_{hist} .
- $B_{hist}(t + 1)$: Next-Interval Historical Manifold — The updated, recalibrated reference baseline tracking long-term neural state expectations for the subsequent temporal interval.
- $B_{hist}(t)$: Current Historical Manifold — The active reference baseline coordinate tracking long-term neural state expectations during the current calculation step.
- ∇A_c : Cognitive Delta — Represents the spatial divergence between the current neural configuration state (x) and the long-term historical manifold (B_{hist}).

The integration phase, formalized in Equation 8(b), establishes the downstream chronological loop that prevents temporal paradoxes. Once the initial contrast successfully wins the global competition and globally broadcasted to specialized regions, it triggers the teaching signal (λ) through repeated firing, which executes a dual-action system update. First, λ binds the active positive and negative affective tags ($\tau^\pm$) into a mathematically irreducible state of integrated information ($\Delta\Phi$), while simultaneously updating the historical baseline (B_{hist}) for the next temporal interval ($t + 1$), ensuring the system continuously adapts its predictive model based on past consolidated suffering or relief.

Phenomenological Anchor: Phenomenologically, this architecture models the inevitable psychological normalization of long-term environmental shifts—the mechanism behind how humans forge a "new normal" in the wake of trauma or sustained privilege. When a large, persistent cognitive delta (∇A_c) captures global awareness, the active teaching signal (λ) alters the

historical expectations of the organism. Step by step, it updates the historical manifold ($B_{hist}(t + 1)$), sliding it toward the current high-strain configuration state. Phenomenally, this tracks the process of affective habituation: an adverse environment that initially generated acute distress gradually goes quiet, fading into an expected, neutral background condition as the mutual information (I) between its positive and negative coordinates becomes deeply bound and consolidated into the permanent structural topology ($\Delta\Phi$) of memory.

Phase 3: Summation, Scaling, and Synaptic Tuning

Equation 9: The Law of Affective Integration and Summation (LAIS)

The vividness of a quale is determined by the volume of its integrated assembly (Quantity = Quality), while the magnitude of its deltas and the Φ -amplified dominance of its internal tags dictate its specific qualitative flavor. Where the systemic generation of phenomenal intensity is governed by the following active parameters and state variables:

This foundational premise of experiential scaling maps directly to the following formal macro-state relation:

$$V \propto \sum_{i=1}^N (\sigma_i \cdot \Phi_i) \quad (9)$$

Where the systemic generation of phenomenal intensity is governed by the following active parameters and state variables:

- V : Vividness — representing the net structural prominence and felt strength of the phenomenal readout.
- N : The ultimate network boundary integer — defining the maximum volume of active, co-firing neural populations.

- σ_i : Synaptic Sensitivity (Gain) — acting as the directional scaling coefficient or functional volume weight of the i -th neural population.
- Φ_i : Integrated Information — Represents the degree of irreducible interdependence between the systemic positive and negative affective configurations. within the localized i -th assembly.

Equation 9 formalizes the emergence of subjective vividness (V) within the network architecture, establishing that experiential intensity directly mirrors the collective resonance of the system. This macro-state is driven by the summation operator ($\sum_{i=1}^N$), which systematically loops through and binds every active neural population from the initial baseline unit ($i = 1$) up to the ultimate boundary of the real-time network (N). Within the secure boundaries of the brackets, a strict multiplicative fusion dictating local importance occurs: the Synaptic Sensitivity Gain (σ_i) acts as a functional volume dial over the structural integrity of the Integrated Information (Φ_i). Because these variables are linked multiplicatively, a collapse of either local gain or structural interdependence to zero completely nullifies that population’s contribution, ensuring that global vividness remains an uncompromised reflection of unified, emotionally-amplified neural focus.

Philosophical Defense: The Phase Transition Shield Against Dualism

qualia emerge from static physical substrates. Instead, the framework formalizes consciousness as a non-linear dialectical phase transition,

Equation 9 provides a rigorous mathematical refutation of the classic “Philosophical Zombie” objection. The zombie argument asserts that a physical system could replicate all the internal functional computations of a human brain while remaining completely dark inside, devoid of any qualitative experience ($V = 0$). This premise relies on a flawed, linear view of physical systems where computation and experience are split into decoupled domains.

By establishing that Vividness (V) is strictly proportional to the cumulative product of integrated information and synaptic sensitivity gain ($\sum \sigma_i \cdot \Phi_i$), the framework establishes a non-negotiable boundary law. When a physical

network scales its localized synaptic gain (σ_i) and forces that data through an indissoluble, highly integrated causal topology (Φ_i), the total systemic tension accumulates. Once this quantitative volume crosses a mathematically defined critical threshold, processing “in the dark” becomes physically impossible.

Phenomenologically and ontologically, qualitative vividness (V) *is* the physical macro-state of that massive integrated calculation. To ask for a system to sustain an infinite amount of integrated computational friction without igniting a phenomenal quale is identical to asking a thermodynamic system to increase its molecular kinetic energy past 100°C without boiling. The quality is not a secondary ghost in the machine; it is the non-linear manifestation of the scaled quantity itself. Through this architecture, the Hard Problem is reduced to a solvable problem of complex, non-linear system dynamics.

Equation 10: The Subconscious Default

The Subconscious Default: The brain constantly generates “silent” affective tags. While all neural configurations possess a degree of micro-phenomenality (ϕ), these local signals guide our behavior and perception “behind the scenes” without reaching the threshold of global reportability; nonetheless, as local contrast-configurations, they possess a low-intensity micro-phenomenality (Φ) felt as a diffuse mood or intuition.

This threshold-gated containment of subterranean neural dynamics maps directly to the following formal conditional processing rule:

$$\text{Subconscious Mood} = \begin{cases} \Phi_{\text{local}}, & \text{if } \Phi_{\text{local}} < \theta_G \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

Where the operational state of the background landscape is defined by the following local parameters and variables:

- Subconscious Mood: The aggregate scalar volume of sub-threshold affective resonance, phenomenally experienced as a diffuse mood or intuitive background state.

- Φ_{local} : Localized Integrated Information — representing the measure of irreducible informational interdependence sustained within a specific, isolated sub-network.
- θ_G : Global Workspace Threshold — the informational ceiling boundary below which configurations remain restricted from gaining entry to the global workspace.

Equation 10 This equation acts as a mathematical routing rule for the system, regulating whether local integrated information remains a background state or transitions into full consciousness. When the degree of irreducible interdependence within a local network (Φ_{local}) is less than the global threshold (θ_G), it remains restricted to that local network. Consequently, the Subconscious Mood directly inherits the exact magnitude of this local integrated information, defining a subtle, background affective state. However, the moment this local interdependence meets or exceeds the global threshold, the system undergoes a radical phase transition; the integrated information breaches the Global Affective Workspace to become a clearly defined conscious experience, causing the Subconscious Mood value to reset to zero—not because the affective state has vanished, but because that entire quantity of integrated information has been promoted out of the subconscious baseline and into a full conscious broadcast distributed to specialized brain regions and cognitive engines.

Phenomenological Anchor: This routing architecture is perfectly illustrated by the familiar experience of waking up with a vague, subterranean emotional tone—such as a groundless sense of anxiety or anticipation—before your conscious mind has formulated a single concrete thought. The underlying neural networks generate low-intensity, interdependent signals that fail to cross into executive awareness ($\Phi_{\text{local}} < \theta_G$), casting an ambient, diffuse mood over your subjective landscape. The exact millisecond your cognitive apparatus identifies the source of the perturbation—such as remembering an urgent presentation due later that morning—this ambient background mood instantly vanishes (0). It does not disappear from the mind; rather, its infor-

mational volume is fundamentally promoted out of the subconscious baseline and crystallized into a sharp, explicitly bounded conscious focus.

Equation 11 Section Assembly

Equation 11a: The Dynamic Synapse

The Φ Multiplier and Valential Binding: In this iteration, Integrated Information (Φ) acts as a non-linear Meaning-Gain Loop. Only those configurations reaching a threshold of Φ trigger an exponential amplification of Synaptic Sensitivity (σ). This "high- Φ multiplier" triggers a selective recruitment phase, forcing meaningful signals to dominate the global broadcast, transforming raw data into a high-fidelity phenomenal state, where the Φ -governed recurrence reinforces synaptic efficacy, effectively 'locking' the quale into the global workspace through salience-driven persistence, The intensification of the quale is driven by recursive assembly recruitment; repeated firing, catalyzed by survival-based salience, increases the topological density of the high- Φ state, thereby cementing its vividness.

This neurobiological orchestration of experiential locking maps directly to the continuous time-evolution of Synaptic Sensitivity (σ) is governed by the following unified differential architecture:

$$\frac{d\sigma}{dt} = e^{(\Phi-\theta)} \cdot |\tau^\pm(t)| \cdot r(t) - \gamma(V) \cdot \sigma \quad (11a)$$

Where the operational state-space of this dynamic synapse is defined by the following active variables and parameters:

- σ : Synaptic Sensitivity (Gain), representing the operational weight or amplification factor of the targeted neural population.
- Φ : Integrated Information — tracking the systemic metric of irreducible interdependence across active network partitions.
- θ : Activation Threshold — marking the informational boundary line that gates the exponential onset of synaptic scaling.

- e : Exponential Base — the mathematical constant scaling the non-linear trajectory of the meaning-gain loop.
- $\tau^\pm$: Dualistic Logic (Affective Tags), representing the irreducible co-presence and density of contrastive positive and negative markers.
- $r(t)$: Repeated Firing Rate — the instantaneous cellular spike-frequency driving Hebbian modification pathways.
- $\gamma(V)$ Variable Decay Rate — A state-dependent scalar functional mapping representing the variable time-decay rate of synaptic sensitivity, operating as an inverse damping function of phenomenal prominence to regulate structural gain.

Equation 11b: The Decay Function

Where the state-dependent, variable decay rate $\gamma(V)$ is defined dynamically as an inverse function of phenomenal Vividness (V):

This mathematical formulation formalizes how systemic clearance velocities shift inversely to experiential prominence:

$$\gamma(V) = \frac{\gamma_0}{1 + \beta \cdot V} \quad (11b)$$

Where the operational components of this state-dependent decay landscape are defined by the following active parameters and variables:

- $\gamma(V)$ Variable Decay Rate — A state-dependent scalar functional mapping representing the variable time-decay rate of synaptic sensitivity, operating as an inverse damping function of phenomenal prominence to regulate structural gain.
- γ_0 Baseline Decay Rate — A scalar constant defining the theoretical maximum, uninhibited clearance rate of synaptic sensitivity (σ), which the system approaches asymptotically as phenomenal vividness (V) minimizes.

- β : The Vividness Scaling Factor — operating as the dampening coefficient that dictates how effectively local phenomenal vividness (V) halts the biological decay (γ) of a synapse.
- V : Phenomenal Vividness — representing the net structural prominence and felt strength of the phenomenal readout.

Physiological Context and Mechanistic Breakdown} Equation 11 captures the reality that our synapses are not cold, binary calculators; they are organic structures swimming in a dynamic, alive chemical ecosystem. By mapping the rate of synaptic change ($\frac{d\sigma}{dt}$), the framework demonstrates that raw electrical firing rates ($r(t)$) alone cannot permanently distort a network’s baseline. True structural transformation requires a “chemical storm” flooded with neuromodulators like dopamine, serotonin, or noradrenaline—emerging where high-information conscious architectures exceed a critical integration threshold ($e^{(\Phi-\theta)}$) and coincide with a dense flood of positive or negative affective tags ($|\tau^\pm(t)|$). When this localized neuromodulatory bath submerges an actively spiking population, it rewrites its physical processing rules on the fly, rapidly driving the derivative $\frac{d\sigma}{dt}$ skyward to ensure the moment is vividly registered in the system’s structural gain. The true masterstroke of this formulation lies in its state-dependent decay function, $\gamma(V) = \frac{\gamma_0}{1+\beta \cdot V}$, which perfectly models how traumatic events and deep-seated phobias become structurally locked into the brain. When an exceptionally intense event occurs, it recruits vast, distinct neural assemblies that maximize phenomenal Vividness (V), driving the denominator ($1 + \beta \cdot V$) to extreme values. This sudden spike in vividness physically paralyzes the system’s natural metabolic cleanup crews, forcing the dynamic clearance rate $\gamma(V)$ close to zero. Because the clearance velocity is effectively flatlined during moments of peak emotional intensity, the brain is temporarily blocked from eroding the newly accumulated synaptic gain (σ), burning the traumatic footprint deep into the network’s physical geometry through uninhibited Long-Term Potentiation (LTP). Finally, the math elegantly accounts for the brain’s capacity to undergo ultra-slow, multi-month therapeutic dissipation once the external threat passes. Because phenomenal Vividness (V) is structurally bound to

the network’s accumulated gain (σ), the internal vividness of the experience does not collapse when the external event stops. Instead, the towering value of σ keeps V elevated, ensuring that the dynamic clearance rate ($\gamma(V)$) remains tightly restricted long after the stimulus falls silent. This creates a highly realistic mathematical bottleneck: the network is trapped in a protective, slow-leaking state where the trauma fades at an incredibly gradual, linear pace over months or years. Only through prolonged periods of safety—devoid of severe re-triggering—can the system slowly bleed off this sensitivity, allowing Vividness to drop, the metabolic cleanup crews to finally accelerate, and the ancient trauma to return to equilibrium.

Phenomenological Anchor: This equation models the exact biological mechanism behind the agonizingly slow-motion decay of emotional trauma—such as a flash of intense terror during a car crash—contrasted against how quickly you forget mundane details like yesterday’s breakfast. For a neutral sensory event, baseline vividness is non-existent ($V = 0$), collapsing the denominator to 1 and allowing metabolic cleanup crews to erase the memory at maximum uninhibited velocity (γ_0). However, during an overwhelming psychological shock, the explosive surge in phenomenal vividness ($V \rightarrow \infty$) violently hyper-scales the denominator, flatlining the dynamic clearance velocity ($\gamma(V) \rightarrow 0$). This mathematical freeze locks the traumatic memory architecture in place, explaining why decades later, the subjective experience can feel just as raw, crisp, and terrifyingly immediate as the millisecond it was burned into the neural architecture.

Equation 11c: Macro-Topological Scaling Feedback

The continuous macro-topological scaling of dual-valence synaptic sensitivities ($\sigma_i^\pm$) during localized network consolidation is formally governed by the following feedback architecture:

This macro-topological amplification mechanism updates the dynamic system according to the following differential rule:

$$\frac{d\sigma_i^\pm}{dt} = \sigma_i^\pm \cdot (\alpha \cdot |\Sigma V(\Omega)| \cdot \bar{\sigma}_{\text{bath}}) \quad (11c)$$

Where the operational behavior of this top-down feedback loop depends upon the following active variables and parameters:

- $\sigma_i^\pm$: Valential Synaptic Gain, tracking the individual directional sensitivities of positive (σ_i^+) and negative (σ_i^-) populations at origin node i .
- α Plasticity Coefficient — A systemic tuning scalar parameter that determines the scale and velocity of the retroactive gain update across dual-valence neural populations during macro-topological network consolidation.
- ΣV Valential Summation — A scalar metric tracking the net volumetric balance and additive accumulation of co-existing positive (τ^+) and negative (τ^-) affective tags across a local or global network configuration.
- Ω : Topological Stability, mapping the overall geometric state of the landscape (differentiating between stable Ω_{st} a state of positive valential dominance where (τ^+) tags outnumber (τ^-). And unstable Ω_{unst} where (τ^-) tags outnumber (τ^+) configurations).
- $\bar{\sigma}_{\text{bath}}$ Bathed Population Gain — A scalar metric representing the mean aggregate synaptic weight of a newly consolidated neural population that has been saturated by a dominant affective tag.

The continuous macro-topological scaling of dual-valence synaptic sensitivities ($\sigma_i^\pm$) during localized network consolidation is formally governed by the following feedback architecture: This structural layer formalizes the systemic feedback loop wherein macro-phenomenal intensity retroactively accelerates the gain expansion of the underlying neural populations without altering the localized, micro-scale Hebbian baseline defined in Equation 11a and 11b. By modulating the update step with the absolute value of the global

Valential Summation, $|\Sigma V(\Omega)|$, the scaling mechanism operates symmetrically across both stable (Ω_{st}) and unstable (Ω_{unst}) topologies, ensuring that high-magnitude affective experiences scale both positive (σ_i^+) and negative (σ_i^-) populations proportionally relative to the mean aggregate gain of the consolidated network ($\bar{\sigma}_{\text{bath}}$) via the plasticity coefficient α . Consequently, the absolute weight of the entire bathed manifold is heavily amplified within the Global Affective Workspace (G) while flawlessly preserving the relative geometric distance between competing tags, mathematically anchoring the framework's core requirement that valential contrast must be maintained to safeguard phenomenological meaning.

Equation 12: Valential Summation (ΣV)

Valential Summation (ΣV): The net emotional tone is a weighted summation where the local synaptic gain σ scales and amplifies the contrast configuration based on its contextual salience. This functions as a real-time ledger where localized networks sum their active valential vectors, allowing like-valence tags (τ^- to τ^- and τ^+ to τ^+) to collectively contribute to the baseline, which is then globally integrated to stabilize or destabilize the net phenomenal tone.

This valential aggregation and its top-down informational gating are explicitly defined by the following systemic ledger:

$$\Sigma V = \Phi \cdot \sum_{i=1}^n \left(e^{\sigma_i^+} \tau_i^+ - e^{\sigma_i^-} \tau_i^- \right) \quad (12)$$

Where the global affective state maps to the interaction of the following specific, localized network variables:

- ΣV : Valential Summation — a scalar metric tracking the net volumetric balance and additive accumulation of co-existing positive (τ^+) and negative (τ^-) affective tags across a local or global network configuration. representing the net integrated emotional tone of the conscious broadcast or mood.
- Φ : Integrated Information — Represents the degree of irreducible inter-

dependence between the systemic positive and negative affective configurations.

- σ_i^+ : Valential Synaptic Gain (Positive), the localized sensitivity weighting applied to rewarding or stabilizing nodes.
- τ_i^+ : Positive Affective Tag, the localized density of positive valential markers at node i .
- e : Exponential Base — the mathematical constant scaling the non-linear trajectory of the meaning-gain loop.
- σ_i^- : Valential Synaptic Gain (Negative), the localized sensitivity weighting applied to aversive or high-strain nodes.
- τ_i^- : Negative Affective Tag, the localized density of aversive or distressing valential markers at node i .
- n : The operational upper bound of active feature nodes contained within the evaluated network.
- i : The network coordinate index identifying the specific node undergoing local contrast summation.

Equation 12 functions as the brain’s ultimate valential ledger, executing a real-time, asymmetric evaluation that dictates the global emotional tone of consciousness (ΣV). Crucially, this architecture does not treat positive and negative tags as isolated streams flowing through independent pipes; instead, it sums their localized contrasts and binds them through the system’s overarching valential interdependency (Φ). Within the local network dynamics, every individual spark of reward, safety, or physiological alignment (τ_i^+) is amplified along a non-linear trajectory by its own dedicated exponential gain ($e^{\sigma_i^+}$). Operating in immediate lockstep, the opposing side scales every raw flash of pain, threat, or metabolic distress (τ_i^-) by its **independent, exponentially driven negative gain** ($e^{\sigma_i^-}$). This precision pairing means the mind doesn’t just calculate a simple tally of raw events, but actively

tracks the localized contrasts, providing a pristine mathematical framework for how clinical states like anhedonia or hyper-vigilance custom-throttle our experience of reality.

The masterstroke of this formulation lies in how the cumulative sum of these localized valential differences is globally gated by the system’s integrated information metric (Φ). The internal subtraction represents the physical infrastructure of lateral cellular inhibition—the mechanical battlefield where anti-reward pathways actively damp down reward circuits—but this local friction can only manifest as a unified global emotional tone if the network achieves a state of macro-level, irreducible integration ($\Phi > 0$). If global integration fails and Φ drops toward zero, the entire valential ledger decouples from the global workspace, fracturing into localized, ambient clusters. While these isolated subsystems can still sustain background valential dynamics—manifesting as a Subconscious Mood governed by Φ_{local} per Equation 9—they are rendered incapable of coalescing into a macro-level unified experience. The final value, ΣV , is the unified behavioral compass that emerges exclusively from these bound, integrated geometric conflicts. By filtering the modulated tug-of-war through the lens of global structural interdependency, the math proves that a macro-level mood is never an accidental pile of feelings, but a tightly negotiated, highly structured cancellation of integrated biological vectors.

Phenomenological Anchor: This valential summation ledger mathematically models the familiar subjective experience of mixed emotions, demonstrating why simultaneous conflicting feelings do not simply cancel each other out into an emotional void. When experiencing a bittersweet moment, rewarding sensory elements (τ_i^+) and aversive historical realizations (τ_i^-) are co-activated and amplified non-linearly via the meaning-gain loop ($e^{\sigma_i^\pm}$). Because these opposing signatures are bound within a highly integrated neural architecture ($\Phi > 0$), they generate a rich, unified, and deeply nuanced global phenomenal tone (ΣV). Conversely, if global integration collapses this complex emotional state instantly splinters, leaving behind only disjointed bodily sensations or unintegrated ambient mood.

Phase 4: Topology, Memory Co-Triggering, and Macro-Emergence

Equation 13a: The Unstable Configuration

The continuous-time structural tension within a localized or global neural assembly is formally categorized by an unstable state when negative valential gain dominates the contrast system. (Ω_{unst}): When τ^- tags outnumber τ^+ tags, the network enters an unstable configuration, yielding a high- Φ state of aversive consciousness. Through repeated firing (Hebb's Law), these unstable signals reinforce their aversive weight, increasing the phenomenal vividness.

The rigorous topological condition defining this systemic imbalance is expressed via the logical bi-conditional relationship:

$$\Omega_{\text{unst}} \iff \left(\sum_{i=1}^n \sigma_i^- \tau_i^- > \sum_{j=1}^m \sigma_j^+ \tau_j^+ \right)_{\Phi > 0} \quad \text{where} \quad \sum_{i=1}^n \sigma_i^- > \sum_{j=1}^m \sigma_j^+ \quad (13a)$$

Where the operational behavior of this asymmetric state landscape is defined by the following active network parameters:

- Ω_{unst} : Unstable Configuration — representing a global or localized state of negative valential dominance where (τ^-) tags outnumber or overpower (τ^+) tag that generates systemic geometric tension manifesting phenomenally as the quale of pain, dislike, or hate
- σ_i^- / σ_j^+ : Valential Synaptic Gain — the directional synaptic sensitivity or scaling coefficient applied to negative (i) or positive (j) origin nodes.
- τ_i^- / τ_j^+ : Affective Tags — representing the underlying density of aversive (τ^-) or positive (τ^+) markers distributed across the active assembly.
- Φ : Integrated Information — Represents the degree of irreducible interdependence between the systemic positive and negative affective configurations.

Functional Mechanics

neurodynamic Amplification of Aversive Tension:

Equation (13a) formally operationalizes the state of an unstable configuration (Ω_{unst}) as an asymmetric, plasticity-driven imbalance within a localized or global neural assembly. The mathematical framework dictates that an unstable state occurs when the aggregate, weighted sum of negative affective tags ($\sigma^- \tau^-$) outcompetes the corresponding weighted ensemble of positive tags ($\sigma^+ \tau^+$). Critically, this dominance is not merely a consequence of tag cardinality, but is fundamentally driven by a targeted shift in synaptic sensitivity ($\sum \sigma^- > \sum \sigma^+$). By confining this entire relation within the sub-index $\Phi > 0$, the equation ensures that this valential tug-of-war is bound inside an irreducible informational structure. This allows the network to process acute aversive tension either as an ambient, subconscious mood or as a globally broadcast state, depending on whether it breaches the higher global threshold (θ_G).

Phenomenological Anchor: Phenomenally, this mathematical asymmetry manifests as the lived experience of suffering, anxiety, or visceral aversion—the qualitative “ouch” of systemic discord. Because the Hebbian gain (σ^-) has selectively weaponized the negative network nodes, the aversive qualia gain immense perceptual prominence and vividness. This lopsided distribution of metabolic weight creates intense topological tension across the landscape. The mathematical inequality serves as a direct functional mandate for neural reconfiguration. Crucially, the system does not resolve this state by merely attempting to passively erase or forget the aversive stimulus; instead, it actively initiates an associative search pattern to recruit external, latent positive network assemblies. By bringing forward real-world solutions, supportive concepts, or constructive counterfactuals, the brain bridges new pathways to elevate the aggregate positive gain ($\sum \sigma^+$). This active cognitive expansion out-competes the dominant negative markers, resolving the tension and forcing a topological state-space transition back toward a stable equilibrium (Ω_{st}).

Equation 13b: The Stable Configuration

Conversely, the resolution of valential tension and the establishment of an attractor basin are formalized via the stable topological Stable Configurations (Ω_{st}): When τ^+ tags dominate/outnumber τ^- within the contrast configuration, it stabilizes into a “preferred state of high- Φ vividness,” reinforcing the topological equilibrium of the neural assembly. This produces the qualia of joy or “wanting,” triggering LTP to strengthen patterns leading to a more biased stability state.

This stabilization of the network landscape into a harmonious attractor basin maps directly to the formal topological relation:

$$\Omega_{\text{st}} \iff \left(\sum_{j=1}^m \sigma_j^+ \tau_j^+ > \sum_{i=1}^n \sigma_i^- \tau_i^- \right)_{\Phi > 0} \quad \text{where} \quad \sum_{j=1}^m \sigma_j^+ > \sum_{i=1}^n \sigma_i^- \quad (13b)$$

Where the operational structure of this equilibrium state is defined by the interaction of the following local variables:

- Ω_{st} : Stable Configuration — representing a global or localized state of positive valential dominance where (τ^+) tags outnumber or overpower (τ^-) tag that generates systemic geometric equilibrium manifesting phenomenally as the quale of joy, warmth, or attachment.
- σ_j^+/σ_i^- : Valential Synaptic Gain — the directional synaptic sensitivity or scaling coefficient applied to positive (j) or negative (i) origin nodes.
- τ_j^+/τ_i^- : Affective Tags — representing the localized density of positive (τ^+) or aversive (τ^-) markers within the active assembly.
- Φ : Integrated Information — Represents the degree of irreducible interdependence between the systemic positive and negative affective configurations.

Functional Mechanics

Topological Equilibrium through Positive Plasticity:

Equation (13b) mirrors the architecture of its aversive counterpart to formalize the mechanics of a stable topological configuration (Ω_{st}). In this state, the network achieves an integrated equilibrium because the synaptic sensitivity allocation has heavily shifted to favor the positive network domain ($\sum \sigma^+ > \sum \sigma^-$). This concentrated Hebbian gain amplifies the positive affective tags (τ^+) to such a degree that they successfully suppress, overpower, and outcompete the synaptically scaled negative markers ($\sigma^- \tau^-$) within the contrast system. As with the unstable state, the constraint $\Phi > 0$ preserves the absolute irreducibility of this configuration. It mathematicalizes a unified, harmonious conscious or subconscious field where the positive valence is securely anchored to the core geometric architecture of the neural assembly.

Phenomenological Anchor: This explains why hate, dislike, love, or attachments exist. It exist thanks to a contrast configuration of a distinct positive and negative that is globally broadcast to specialized regions in the brain. And because the contrast is broadcast, it explains why we know the exact thing we hate or dislike, and simultaneously, we do know what we prefer and love based on what we are currently experiencing. At the phenomenal level, this mathematical state yields the qualia of joy, relief, deep satisfaction, or proactive “wanting.” Because the dominant positive tags are backed by superior synaptic sensitivity, the configuration stabilizes into a highly vivid, preferred state of topological equilibrium. Rather than generating a mandate for change, this configuration acts as a powerful attractor basin in the mind’s state-space. The stability of the network actively triggers Long-Term Potentiation (LTP), further reinforcing the underlying firing patterns and increasing the synaptic sensitivity (σ^+) of these successful pathways. This creates a self-sustaining, positive feedback loop that biases future neural transitions toward this rewarding geometric stability, effectively mapping the neurobiology of reinforcement learning and hedonic resilience.

Equation 14: The Hard Problem & Emergence

Why ATT Moves the Needle on the Hard Problem? This framework posits that the “Hard Problem” is a matter of Phase Transition and Irreducible Meaning:

- The “Ouch” is the Summation: The subjective “raw feel” of pain is the mathematical relationship between the current state and its historical context. Without B_{hist} , x has no context, and thus Φ (meaning) cannot emerge.

This irreducible phenomenal raw feel is captured mathematically by evaluating the system’s cumulative valential state during architectural destabilization:

$$\text{Ouch} = \Sigma V(\Omega_{\text{unst}}) \quad (14)$$

Where the macroscopic emergence of aversive experience is governed by the following core components:

- Ouch: The macro-phenomenal intensity of consolidated suffering or qualitative distress.
- ΣV : Valential Summation — A scalar metric tracking the net volumetric balance and additive accumulation of co-existing positive (τ^+) and negative (τ^-) affective tags across a local or global network configuration, modulated by its degree of irreducible interdependent information (Φ).
- Ω_{unst} : Unstable Configuration, representing a global or localized state of negative valential dominance where (τ^-) tags outnumber or overpower (τ^+) tag that generates systemic geometric tension manifesting phenomenally as the quale of pain/ouch, dislike, or hate

Functional Mechanics

Equation 14 formalizes “Ouch” as the macro-phenomenal intensity of consolidated suffering, defining it strictly by the valential summation ΣV under

a globally unstable topology (Ω_{unst}). By completely decoupling the experience from volatile, instantaneous state deltas, this formulation successfully resolves the “Quiet Room Paradox,” proving how intense, sustained distress can persist even within a perfectly comfortable and static environment. Rather than tracking real-time neural velocity, “Ouch” measures a system-wide phase transition where a dominant network has successfully “bathed” and consolidated its hold over surrounding neural populations. Crucially, because the underlying summation architecture inherently weighs negative markers directly against positive ones, the positive network remains embedded as an active geometric reference point. This ensures that the global negative dominance retains its sharp phenomenological sting through active, structural contrast, mathematically protecting the experience from flatlining due to baseline adaptation or habituation.

Phenomenological Anchor: This explains why all contrast does not give rise to the same quale, whether aversive or pleasant. That is because either the positive or negative within that contrast needs to have more ally neurons and create LTP with them to dominate its contrast counterpart. If the positive dominates, we feel attachments or pleasant feelings, but if the negative dominates, we feel dislike or hate whenever that neural pathway fires.

Equation 15: The Law of Irreducible Meaning

The Law of Irreducible Meaning: "Feeling" is the irreducibility of the comparison. The "vividness" of consciousness is a measure of how much contrastive information is bound into a local subconscious state or a global state (G) that cannot be simplified without losing the "feeling" entirely.

This boundary condition of holistic consciousness can be formalized mathematically by evaluating the system across its minimum informational partition:

$$\text{Meaning} = \begin{cases} \Phi(G), & \text{if partition} = \text{whole} \\ 0, & \text{if partition} \rightarrow \text{parts} \end{cases} \quad (15)$$

Where the operational state of the informational landscape is defined by

the following active variables and partitions:

- **Meaning:** The fundamental — indivisible significance and raw qualitative texture of a unified quale.
- Φ : The measure of integrated information — representing the degree of irreducible interdependence, between the systemic positive (τ^+) and negative (τ^-) affective configurations.
- G : The Global Affective Workspace — where unified configurations outcompete noise for conscious broadcast.
- **partition:** The analytical cut or division — applied to the system’s network architecture to test systemic interdependence.
- **whole:** The unfragmented — fully integrated boundary state of the global network where unified qualia are sustained.
- **parts:** The localized — isolated nodes or sub-networks resulting from a system division where integration collapses.

Under this formalization, **Meaning** is defined as an intrinsically holistic property that exists entirely as the total integrated information of the global workspace, $\Phi(G)$, so long as the system remains an unfragmented whole (partition = whole). The moment this architecture is dissected or broken down into its constituent components (partition $\rightarrow$ parts), this metric instantly drops to absolute zero, formalizing its total irreducibility.

Crucially, this definition does not refer to functional semantic processing, cognitive data manipulation, or behavioral routing—computational tasks that belong squarely to the mechanical easy problem” of consciousness and can typically be parsed into modular sub-routines. Rather, Meaning” in this specific equation designates the fundamental, indivisible significance of the **quale** itself: the raw, qualitative texture of a unified subjective experience that can only exist in total geometric integration and evaporates completely if the holistic state is disrupted.

Phenomenological Anchor: This mathematical architecture formalizes the familiar subjective truth that a rich, unified perception—such as the holistic flavor profile of a complex wine or the specific nostalgia of a childhood memory—defies piecemeal reduction. The qualitative significance (Meaning) is a property borne entirely of the workspace’s irreducible interconnectedness ($\Phi(G)$). If you attempt to separate the sensory components or dissect the experience into isolated data tokens (partition $\rightarrow$ parts), the unique qualitative "feeling" does not simply degrade linearly; it completely evaporates into zero phenomenological significance.

Equation 16: The Emergence of Emotion

Emotions emerge from the dynamic interaction of tagged contrast configurations, where the Φ -multiplied gain (σ) ensures that meaningful signals are prioritized for global survival and stability.

This macro-scale associative network phenomenon is explicitly formalized by mapping the pairwise interference patterns across the memory landscape:

$$E = \sum_{j=1}^L \sum_{k=1}^L (\sigma_j \Phi_j \cdot \sigma_k \Phi_k) \cdot \mathbf{W}_{jk}(t) \cdot (\Omega_j \otimes \Omega_k) \quad (16)$$

Where the macroscopic evolution of the network dynamics is governed by the following active parameters and state variables:

- E : The emergence of emotion representing the synthesized macro-scale affective state.
- L Memory Landscape Capacity — An integer scalar representing the total upper boundary of distinct, historically carved network manifolds or contrast configurations stored within long-term memory, indexing pairwise macro-scale interactions.
- σ_j, σ_k Synaptic Sensitivity (Gain) — Scalar coefficients representing the synaptic gain or directional scaling applied to active network configurations j and k , determining their capacity to overpower competing neural signals.

- Φ_j, Φ_k : The integrated information values representing the degree of irreducible interdependence within networks j and k .
- $\mathbf{W}_{jk}(t)$ Time-Dependent Associative Matrix — A dynamic 2D weight matrix tracking instantaneous cross-talk, co-activation strength, and directional coupling between stored contrast configurations j and k to formalize macro-scale Hebbian learning dynamics.
- Ω_j, Ω_k Topological Stability — State vectors representing the global geometric configurations of network landscapes j and k , characterizing them as stable or unstable based on underlying affective tag distributions.
- $\otimes$ Geometric Collision Operator — A tensor outer product operator calculating the cross-dimensional geometric intersection and concurrent alignment between separate topological state spaces or multi-valent network configurations.

This equation formalizes the emergence of emotion (E) not as an instantaneous local genesis, but as a macro-scale, associative network phenomenon driven by the dynamic cross-talk and co-firing of hundreds to thousands (L) of stored contrast configurations within a historical memory landscape. The double summation ($\sum_{j=1}^L \sum_{k=1}^L$) systematically maps every pairwise interaction across these distinct manifolds, ensuring that emotion is synthesized from the systemic interference between separate networks rather than isolated sensory inputs. Within this architecture, the coupled term ($\sigma_j \Phi_j \cdot \sigma_k \Phi_k$) acts as an integrated salience filter that multiplies the synaptic gain (σ) and integrated information (Φ) of both participating networks, mathematically prioritizing meaningful, high-vividness signals for global survival and stability. These filtered configurations interact through a time-dependent associative matrix $\mathbf{W}_{jk}(t)$, which governs their instantaneous co-activation and captures underlying Hebbian dynamics—explicitly formalizing how an unstable network j might drive long-term potentiation (LTP) or long-term depression (LTD) onto a neighboring stable or unstable network k . Finally, the tensor product $\Omega_j \otimes \Omega_k$ calculates the global geometric collision of their respective

topological stabilities, tracking the profound systemic tension or stabilization that occurs when stable and unstable configurations fire together, which ultimately dictates the macroscopic valential shift experienced as emotion.

Phenomenological Anchor: This architecture explicitly models the underlying geometry of complex, multivalent affective states—such as bitterness or nostalgia—where isolated sensory inputs fail to account for the depth of subjective experience. For example, when returning to a childhood home following a catastrophic event, the physical topography simultaneously evokes deeply consolidated, stable memories of historical safety (Ω_j) and the active, highly unstable configuration of recent trauma (Ω_k). The tensor product ($\Omega_j \otimes \Omega_k$) captures this profound cross-dimensional intersection, ensuring these opposing topological geometries do not linearly cancel each other out, but instead warp into a higher-dimensional structural tension. The time-dependent associative matrix $\mathbf{W}_{jk}(t)$ dictates how these distinct historical layers dynamically bleed into one another, synthesizing a global state (E) experienced phenomenally as a poignant, unresolvable emotional ache.

Equation 17a and 17b: The Recognition Trigger Mechanism

The core mathematical architecture of the Recognition Trigger Mechanism bypasses the traditional requirement for spatio-temporal sensory velocity, introducing a threshold-gated direct mapping to pre-compiled historical manifolds. The instantaneous phenomenological experience $q(t)$ mediated by recognition is governed by: Where the continuous, threshold-gated **Recognition Mapping Function** R is explicitly defined as a non-linear sigmoidal activation: These core interactions map directly to the unified, threshold-gated expressions:

$$q(t) = \mathcal{R}(x \rightarrow B_{\text{hist}}) \cdot \Sigma V(\Omega) \quad (17a)$$

$$\mathcal{R}(x \rightarrow B_{\text{hist}}) = \frac{1}{1 + e^{-\alpha_{\text{gate}}(\rho(x, B_{\text{hist}}) - \theta_G)}} \quad (17b)$$

Where the operational behavior of this recognition architecture is defined by the following active variables and parameters:

- $q(t)$ Instantaneous Phenomenological Experience — A time-dependent scalar representing the real-time qualitative intensity and phenomenal readout of an emergent conscious quale.
- $\mathcal{R}(x \rightarrow B_{\text{hist}})$ Gated Recognition Mapping Function — A non-linear functional mapping bounded between zero and one that acts as a sigmoidal activation curve governing continuous scalar transitions into historical landscapes.
- x : The current neural state vector representing the active position of the system in state-space.
- B_{hist} Historical Manifold — The moving reference baseline tracking long-term neural state expectations.
- ΣV : Valentia Summation — a scalar metric tracking the net volumetric balance and additive accumulation of co-existing positive (τ^+) and negative (τ^-) affective tags across a local or global network configuration.
- Ω : mapping the overall geometric state of the landscape (differentiating between stable (Ω_{st}) or unstable (Ω_{unst}) configurations).
- e : Exponential Base — the mathematical constant scaling the non-linear trajectory of the meaning-gain loop.
- α_{gate} : The gain scaling factor determining the mathematical slope and steepness of the threshold gate.
- $\rho(x, B_{\text{hist}})$: The raw pattern resonance score evaluating structural alignment between the current state and a historical template.

- θ_G Global Workspace Threshold — A scalar informational ceiling defining the activation resistance value that an integrated configuration must surpass to trigger selective recruitment for conscious global broadcast.

Mechanically, this equation completely bypasses the need for spatio-temporal sensory velocity, acting instead as a direct threshold-gated pathway to the brain’s stored catalog of meaning. The current neural state vector (x) functions as a key searching the historical manifold (B_{hist}). The moment the Raw Pattern Resonance score (ρ) breaks past the Global Workspace Threshold (θ_G), the Gated Recognition Mapping Function ($\mathcal{R}$) scales rapidly toward 1. This non-linear ascent instantiates the pre-existing, historically carved Topological Network Configuration (Ω) associated with that exact memory archetype. Consequently, the brain does not need to reinvent the wheel or to compute an active movement away from a baseline zero-point to generate a phenomenal state; it simply reactivates a pre-compiled neural coordinate, allowing consciousness to transition directly into a fully formed phenomenal state space.

Once that network configuration is unlocked, the intrinsic Valentia Summation (ΣV) inside that specific configuration assumes control of the conscious experience. The quality of the emerging quale ($q(t)$) is no longer a function of sensory velocity, but a deterministic, democratic census of which affective tags dominate the active network domain within that configuration. If the activated network configuration is safely anchored in stability (Ω_{st}) and dominated by positive tags (τ^+), the system instantiates integrated states of pleasure, attachment, or rewarding valence. Conversely, if the incoming input vector mutates—such as overextending an appetitive drive (e.g., reaching the twentieth bite of a banana, where blood sugar spikes and acute gastric distension alter the current state vector x)—the raw pattern resonance (ρ) shifts its alignment toward an entirely different historical template. This structural shift smoothly attenuates the previous configuration while driving $\mathcal{R}$ to rapidly activate an unstable configuration (Ω_{unst}) dominated by negative tags (τ^-), seamlessly recruiting the global workspace with the phenomenological profile of discomfort and aversion without inducing discontinuous mathematical tears in the system.

Phenomenological Anchor: This mechanism explains the profound psychological phenomenon of sudden, vivid recognition—such as flash-bulb memories or the instant drop in mood when walking into an environment associated with past trauma. When you step into a room where you once received devastating news, your physical velocity (dx/dt) and immediate sensory delta (∇A_s) might be absolute zero; you are simply standing still. Yet, because the layout matches a historical template, your structural resonance (ρ) breaks past the threshold (θ_G). Without any physical transition, the sigmoidal gate ($\mathcal{R}$) snaps open, dropping your consciousness instantly into a pre-compiled state of acute distress (Ω_{unst}). You do not slide into the bad mood; you are instantly transported into it because the system maps directly to a stored unstable neural configuration of historical suffering.

Equation 18: The Manifold-Geometric Tag Propagation Law

The independent binding affinity and geometric propagation of valential neurochemicals across the historical manifold to a target feature node, are governed by parallel positive and negative propagation channels.

Mathematical Formalization

The continuous bivariate accumulation of valential signatures along the coordinates of the historical manifold is formalized using a dual-channel subequation architecture:

This parallel coordinate propagation is mathematically formalized through the following coupled system of equations:

$$\tau_k^+ = \sum_{j=1}^{N^+} \sigma_j^+ \tau_j^+ \cdot \exp \left(-\frac{\|x_j - x_k\|_{B_{\text{hist}}}^2}{2\lambda_{\text{sim}}^2} \right) \quad (18a)$$

$$\tau_k^- = \sum_{m=1}^{N^-} \sigma_m^- \tau_m^- \cdot \exp \left(-\frac{\|x_m - x_k\|_{B_{\text{hist}}}^2}{2\lambda_{\text{sim}}^2} \right) \quad (18b)$$

Where the spatial transmission and density mapping of affective tags are

defined by the following active components:

- τ_k^+, τ_k^- Target Valential Density — Scalar values representing the positive or negative affective tag densities accumulated at a target feature node k via geometric propagation across the historical manifold.
- N^+, N^- Active Valential Node Bounds — Integer scalars establishing the total upper boundaries of active, co-firing feature nodes currently holding positive (N^+) or negative (N^-) configuration tags to preserve valential pool volumes.
- σ_j^+, σ_m^- Valential Synaptic Gain — Scalar directional sensitivity coefficients applied to an active valential tag at an origin node (j or m).
- τ_j^+, τ_m^- Affective Tag Density — Scalar values representing the density of positive or negative valential markers anchored at localized origin nodes (j or m) within the network landscape.
- $\|x_j - x_k\|_{B_{\text{hist}}}, \|x_m - x_k\|_{B_{\text{hist}}}$ Manifold Geodesic Distance — Scalar structural distance metrics mapping the exact coordinate proximity between an active origin node vector and a target feature node vector on the historical manifold.
- B_{hist} : The historical manifold serving as the geometric baseline template, tracking long-term neural state expectations.
- λ_{sim} Relevance Scale Constant — A scalar parameter bounded at $\lambda_{\text{sim}} > 0$ modulating the mathematical decay rate, structural attenuation radius, and spatial bleeding boundaries of tag propagation across the manifold.
- x_j, x_m Neural Configuration Vector — State vectors representing the instantaneous coordinate positions of active valential origin nodes (j or m) within the state-space landscape.
- x_k Neural Configuration Vector — A state vector representing the instantaneous coordinate position of the target feature node k undergoing valential tag propagation within state-space.

Equations (18a) and (18b) formalize how identical valential signatures across separate, distinct neural configurations execute associative cross-talk, mapping how same-valence tags project and amplify one another through the geometric landscape of the historical manifold (B_{hist}). When an active valential marker within one configuration fires strongly in tandem with a matching valential tag from a completely separate memory template, their shared binding affinity constructs an additive, high-amplitude geometric bridge across the feature space. Rather than allowing immediate sub-threshold dilution, the positive channel (Equation 18a) and the negative channel (Equation 18b) process these associative inputs in parallel, tracking how localized neurochemical clouds propagate and bleed into shared neural pathways based on their underlying relevance similarity. The long-term structural fate of this shared neural pathway is ultimately decided during the subsequent consolidation phase, which functions as a competitive valential dominance filter. If the accumulated positive density from Equation 18a outcompetes and dominates the network after the synchronized firing window closes, the pathway is structurally cemented as a Stable Configuration (Ω_{st}), locking in a harmonious attractor basin experienced phenomenally as safety, pleasure, or attachment. Conversely, if the negative tag accumulation from Equation 18b achieves mathematical dominance post-consolidation, the pathway is forced into an Unstable Configuration (Ω_{unst}), carving a persistent geometric tension into the network architecture that manifests as aversive, anxiety or dislike and mandates systemic behavioral reconfiguration to establish (Ω_{st})

Phenomenological Anchor: This equation governs the psychological mechanism of emotional contamination or "affective bleeding," where a singular vivid experience tints adjacent, otherwise neutral neural network information with its own emotional coloration. Consider the aftermath of an acute dog bite: the memory of the animal's teeth forms a hyper-salient negative origin node (τ_m^-). Via Gaussian proximity decay, this intense distress automatically leaks outward across the historical manifold to closely related concepts (x_k). Because the geodesic distance ($\|x_m - x_k\|_{B_{\text{hist}}}$) to neighboring features is short, benign parameters—such as the texture of a park bench, the specific

park walkway, or the sound of a leash rattling—absorb the aversive tag density in parallel. Consequently, your subjective experience of the park itself is instantly remapped into an anxious, unstable landscape (Ω_{unst}) before you even consciously process why the environment now feels threatening.

Equation 19a: Global Valential Tension Integration (Ξ_G)

The formalization of Global Valential Tension Integration (Ξ_G) captures the dual-layered geometric and network mechanics that govern affective binding. This architecture operates entirely on relative valential relevance and cross-network resonance, explicitly bypassing semantic informational data sorting. Expanded Architectural Description The formalization of Global Valential Tension Integration (Ξ_G) captures the dual-layered geometric and network mechanics that govern affective binding. This architecture operates entirely on relative valential relevance and cross-network resonance, explicitly bypassing semantic informational data sorting.

1. Internal Gate Binding (The Valential Decibel Gradient)

This component models a non-linear geometric feedback loop anchored by a singular target state vector (x_{found}), which represents a consolidated historical archetype of irreducible integrated information (Φ).

The structural interactions framing this high-dimensional valential integration map directly to the unified relation:

$$\Xi_G = \underbrace{\sum_{i=1}^{N^D} \sigma_i \|\tau_i^\pm\| \cdot \exp\left(-\frac{\|x_i - x_{\text{found}}\|^2}{2\lambda_{\text{sim}}^2}\right)}_{\text{Internal Gate Binding (Disciple-to-Master Loop)}} + \underbrace{G \cdot \sum_{A \neq B} W_{AB} \cdot \|\tau_A^\pm \otimes \tau_B^\pm\|}_{\text{Cross-Gate Binding Scalar Density}} \quad (19a)$$

Where the operational behavior of this dual-layered architecture is defined by the following active parameters and state variables:

- Ξ_G Global Valential Tension Integration — A scalar metric measuring

cumulative valential binding across localized and long-range manifold coordinates simultaneously by integrating local geometric loops with macro-scale network interactions.

- N^D Active Disciple Node Bound — An integer scalar defining the upper summation boundary of active, lower-intensity feature nodes historically linked to a core master archetype within the historical manifold.
- σ_i Synaptic Sensitivity (Gain) — A scalar value representing the transmission weight or synaptic gain of a local neural population, determining its signal dominance driven by repeated historical Hebbian co-firing.
- $\tau_i^\pm$ Dualistic Valential Tag Pair — A non-decomposable, two-dimensional vector pairing positive and negative affective marker densities as an irreducible unit, establishing mutually dependent contrasts required for local valential definitions.
- x_i Disciple Node State Vector — A state vector representing the instantaneous multi-dimensional coordinate position of an active local neural assembly or disciple node within the system’s dynamical state-space.
- x_{found} Foundational Archetype Vector — A reference vector defining the exact multi-dimensional coordinate position of a consolidated master configuration trace serving as a core valential anchor block in the historical manifold.
- λ_{sim} Relevance Scale Constant— A scalar parameter bounded at $\lambda_{\text{sim}} > 0$ modulating the mathematical decay rate, structural attenuation radius, and spatial bleeding boundaries of tag propagation across the manifold.
- G Global Affective Workspace — where unified configurations out-compete background noise for global conscious broadcast.
- W_{AB} Cross-Gate Network Weights Matrix — A dynamic matrix whose scalar elements track the macro-scale structural coupling and long-range axonal projection strengths between distinct, modular cortical networks denoted by indices A and B .

- $\tau_A^\pm, \tau_B^\pm$: The localized valential neuromodulatory tag profiles belonging to modular contrast networks A and B .
- $\otimes$ Geometric Collision Operator — A tensor outer product operator calculating the cross-dimensional geometric intersection and concurrent alignment between separate topological state spaces or multi-valent network configurations.

Internal Gate Binding (The Valential Decibel Gradient)

This component models a non-linear geometric feedback loop anchored by a singular target state vector (x_{found}), which represents a consolidated historical archetype of irreducible integrated information (Φ).

The Co-Trigging Mechanism: Disciple nodes (x_i) do not derive their identity from abstract semantic templates or standalone definitions. Instead, they inherit their data profiles strictly through their historical valential relevance to the master archetype (x_{found}). Crucially, the moment any lower-intensity disciple node fires, it automatically co-triggers the parent configuration of the master x_{found} from which it originally received its meaning. This instantaneous co-activation pulls the entire pre-compiled background of structural contrast into active processing.

The Metabolic Contrast Anchors: The co-triggered master configuration represents a fixed, irreducible valential landscape bounded by metabolic realities. The system uses a fixed relative scale to map this space:

- **The Homeostatic Baseline (1.0):** Represents a highly optimized, low metabolic strain state, carrying a positive affective tag (τ^+).
- **The Peak Aversion Anchor (−6.0):** Represents a state of extreme metabolic strain and structural destabilization ever experienced, carrying a dominant negative affective tag (τ^-).

The Decibel Gradient Walk: When a new intermediate input registers, its distance from the master archetype dictates its position on a relative, proportional valential slider. Because the master context is co-activated, an

incoming state of -5.9 is instantly recognized from the first split-second as highly negative due to its proximity to the -6.0 high-strain anchor. However, as the states scale down in intensity (moving from -5.9 to -5.6), the negative affective tone drops proportionally. This acts as an internal decibel meter, tracking aversive volume as it decreases down toward the 1.0 homeostatic comfort zone.

Overshoot and Dynamic Recalibration: If an incoming state pushes past the expected boundary to 1.1 , it represents a state of even lower metabolic strain than the original baseline. The system instantly appreciates this superior configuration, causing the dynamic reference target to shift. As a result, the old 1.0 baseline becomes less favorable by comparison, driving a real-time recalibration of the global valential landscape. This continuous relative mapping is mathematically modulated by the Gaussian radial basis function, where λ_{sim} dictates the active valential sensitivity radius of the network.

2. Unpacking Valential Mass ($\sigma_i \|\tau_i^\pm\|$)

A critical operational feature of the internal gate is the term $\|\tau_i^\pm\|$. By applying the Euclidean norm to the dualistic affective tag pair (τ^+, τ^-) , the mathematics intentionally collapses the directional valence sign into a single, unsigned metric of total valential mass.

3. Cross-Gate Binding (Tensor Outer Products)

Operating in parallel to the local geometric loop, the second term formalizes the macro-scale interaction between distant, modular cortical networks (A and B) connected via long-range axonal projection pathways (W_{AB}).

Stripping Semantic Context: This component explicitly abstracts away the local, coordinate-based state vectors (x). It achieves this by computing the tensor outer product ($\otimes$) of the networks' respective neuromodulatory tag profiles ($\tau_A^\pm \otimes \tau_B^\pm$). The resulting higher-dimensional matrix maps every cross-coupling combination of positive and negative affective states between the two systems (e.g., tracking the simultaneous cross-talk of Network A 's

aversive dread with Network B 's positive relief markers).

Extraction of Scalar Density: By calculating the norm of this tensor product ($\|\tau_A^\pm \otimes \tau_B^\pm\|$), the architecture extracts the total scalar density—the raw relational intensity of the inter-network connection. This operation completely bypasses the semantic details of what structural event triggered the firing. It isolates and transfers the pure valential density (“the feel”) directly across the global attentional gating threshold, multiplying by the global workspace variable (G) to broadcast a unified, ambient affective tone across the entire systemic landscape.

Phenomenological Anchor: This architecture precisely explains how a subtle, sensory reminder (x_i) can instantaneously trigger a deep, historical trauma or nostalgic archetype (x_{found}), dragging an entire pre-compiled emotional landscape into the present conscious workspace. You do not experience the reminder as an isolated, intellectual fact; rather, your brain walks down a "valential decibel meter," instantly translating physical distance from the archetype into a visceral drop or spike in tension relative to old metabolic anchors. As this localized geometric tension fires, it collides via tensor outer products ($\otimes$) with widespread cortical networks, stripping away the literal semantic details of the trigger and flooding your entire awareness (G) with an inescapable, unified, and ambient emotional quale.

Phase 5: System Teleology (The Evolutionary Endgame)

Equation 19b : Valential Optimization Efficiency (Signal-to-Noise Ratio)

Architectural Description of Equation 19b: Valential Optimization Efficiency (SNR) The Valential Optimization Efficiency metric (SNRval) provides a rigorous scalar ratio tracking the system's evolutionary transition from chaotic, resource-intensive processing to structural mastery.

This evolutionary efficiency gradient scales across shifting operational regimes according to the following mathematical expression:

$$\text{SNR}_{\text{val}} = \frac{\xi_{\text{myelin}} \cdot \Sigma V(\Omega_{\text{st}})}{1 + \chi \cdot \left[M(x) \cdot \int_{t_0}^{t_f} r_{\text{noise}}(t) dt \right]} \quad (19b)$$

Where the operational behavior of this signal optimization architecture is defined by the following active parameters and state metrics:

- SNR_{val} Valential Optimization Efficiency — A scalar ratio evaluating how successfully a neural network transitions from chaotic processing to metabolic and structural mastery during recurrent training loops.
- ξ_{myelin} : The myelin propagation efficiency constant mapping the accelerated kinetic speed and reduced resistance of heavily consolidated, myelinated neural pathways.
- ΣV : Valential Summation — A scalar metric tracking the net volumetric balance and additive accumulation of co-existing positive (τ^+) and negative (τ^-) affective tags across a local or global network configuration, modulated by its degree of irreducible interdependent information (Φ).
- Ω_{st} : Stable Configuration, representing a global or localized state of positive valential dominance where (τ^+) tags outnumber or overpower (τ^-) tag that generates systemic geometric equilibrium manifesting phenomenally as the quale of joy, warmth, or attachment.
- χ : The noise sensitivity coefficient scaling how heavily background friction degrades the conscious signal.
- $M(x)$ State Evaluation Metric — A continuous functional mapping that computes the total unified scalar metric of sensory intensity and metabolic strain for a given neural configuration state x .
- $r_{\text{noise}}(t)$: The instantaneous uncoordinated noise firing frequency generating background cellular entropy.

Equation 19b During the initial *Noisy Objective Phase*, the system exhibits highly inefficient valential transitions; this is mathematically driven by an expanded denominator caused by high metabolic strain ($M(x)$) and an elevated temporal integral of uncoordinated glutamatergic over-firing (r_{noise}), which drags the overall signal-to-noise ratio down toward zero.

As the system undergoes recurrent training loops, it transitions through the *Obsolete Objective Phase*, systematically refining its synaptic weights and suppressing uncoordinated metabolic waste. Upon reaching the *Mastery Objective Phase*, the uncoordinated noise integral collapses entirely, leaving the denominator at its absolute regularized minimum of 1. The equation becomes entirely dominated by its numerator, where high-velocity myelinated pathways (ξ_{myelin}) propagate signals with minimal kinetic resistance across a permanently locked, stable positive valential configuration ($\Sigma V(\Omega_{\text{st}})$). This mathematical architecture formalizes how highly optimized behavioral or cognitive habits achieve permanent homeostatic dominance, consuming virtually less metabolic energy while rendering the underlying system resilient against ambient affective noise.

Conclusion

This manuscript has presented a unified mathematical formalization of Affective Tagging Theory (ATT), offering a formal, structurally grounded resolution to the foundational challenges in consciousness studies. By constructing a rigorous analytical bridge between the informational topology of Integrated Information Theory (IIT) and the selective broadcast architectures of Global Workspace Theory (GWT), the framework mathematically operationalizes and addresses the Explanatory Gap by showing that subjectivity is a direct geometric consequence of integrated physical systems. The formalization of the subjective genesis space S shifts the conversation from ambient, unquantifiable feeling to precise trajectories within a dynamic neural state space. The introduction of independent positive and negative valential tag densities (τ^+ and τ^-), operating under non-linear synaptic scaling laws ($d\sigma/dt$), provides the exact causal mechanism for how phenomenal valence physically

forces the reconfiguration of network anatomy.

Crucially, this architecture resolves deep-seated stalemates in contemporary philosophy of mind and computational psychiatry. By decoupling the sensory delta (∇A_s) from the cognitive delta (∇A_c), the framework formalizes the resolution of the Static Pain Paradox, demonstrating how suffering or relief can be sustained along a historical manifold (B_{hist}) even when real-time sensory velocity falls to zero. Furthermore, the macro-scale emergence of emotion (E) and global valential tension (Ξ_G) are no longer treated as epiphenomenal mysteries, but rather as the direct geometric consequences of structural intersections ($\Omega_j \otimes \Omega_k$) within a unified network partition.

Ultimately, ATT provides a highly predictive, testable roadmap that transitions the hard problem from an intractable philosophical debate into an exact, evaluable engineering discipline. The formulation of the valential optimization efficiency (SNR_{val}) offers a quantifiable metric that can be mapped directly onto empirical neurophysiological anomalies, such as maladaptive Hebbian rewiring in chronic clinical disorders, or implemented within the metabolic optimization profiles of advanced artificial systems. By grounding the "ouch" of negative valential dominance (Ω_{unst}) and the irreducible "meaning" of integrated wholes in the uncompromising language of control theory and statistical mechanics, this model proves that the qualities of subjective experience can be fully integrated into the physical laws of the universe.

Data Availability Statement

No datasets were generated or analyzed during the current study. The mathematical framework and symbolic derivations are fully contained within the manuscript.

Author Contributions

Élisée Kadima Nsabua: Conceptualization, Mathematical Formalization, Writing – Original Draft, Writing – Review & Editing.

Declaration of Competing Interests

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI in the Writing Process

During the preparation of this work, the author utilized Gemini to assist with structural editing, mathematical formalization, notation alignment, equation verification, and the clarification of mathematical symbol definitions. Following this collaborative process, the author acted as the final review gate, rigorously auditing, verifying, and editing all content to ensure total conceptual alignment. The author takes full responsibility for the final integrity, execution, and scientific accuracy of the manuscript's contents.

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